HM811 Natural Capital Extent Accounts in R

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Introduction

Understanding changes in habitat types and their use over time is crucial to assessing ecolsystem health. This document demonstrates the creation of extent accounts as a possible tool for ecological monitoring.

Maps

The below maps are taken from Hazelwood farm, courtesy of the For-ES project at Trinity College Dublin `https://www.for-es.ie/`.

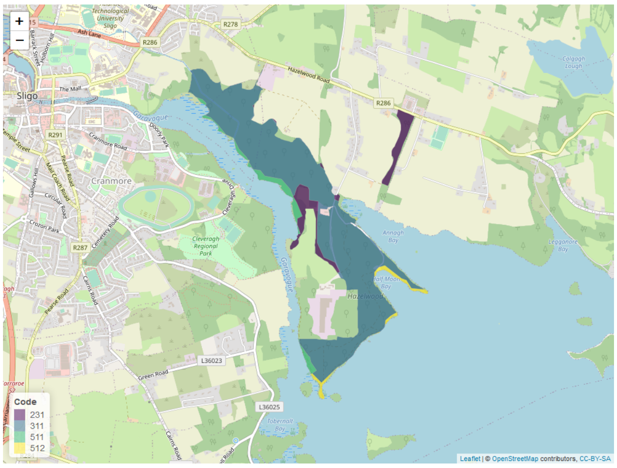

Figure 1: Site in 2000, m_0

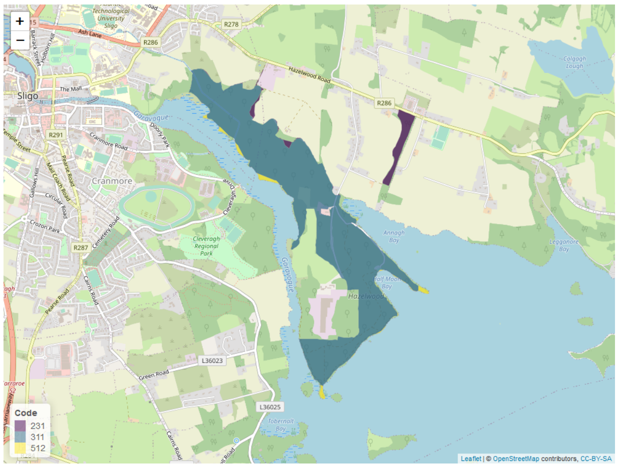

Figure 2: Site in 2018, m_1

Notice that code **511** (water courses), which is in the site in 2000, is absent from the site in 2018 and is replaced wirth 311 (broad-leaved forest). We also norice some small amounts of **231** (pasture) has been reintroduced in the north east. Some more investigation is needed to figure out the habitat dynamics.

Extent accounts

Given two shapefiles like those displayed in Fig 1 and Fig 2, we want to calculate the changes in each group between the first and second timepoint.

Suppose we have opening and closing maps m_0 and m_1 respectively. Their extent is calculated this way:

- For the opening area of each group $g \in \{231, 311, 511, 512\}$ we calculate the opening area $A_{\text{opening},g}$ to be

$$A_{\text{opening},g} = \sum_{\substack{j \in m_0 \\ j \text{ in group } g}} \text{Area}(j) \quad \text{for all polygons } j \text{ and groups } g$$

  and closing area $A_{\text{closing},g}$

$$A_{\text{closing},g} = \sum_{\substack{k \in m_1 \\ k \text{ in group } g}} \text{Area}(k) \quad \text{for all polygons } k \text{ and groups } g$$

  Where Area is a function calculating the area of a polygon (in Hectares).

- The intersection B_g between m_0 and m_1 (for each group) gives the amount unchanged (if any) across the time period

$$B_g = \sum_{\substack{j \in m_0 \\ k \text{ in group } g}} \left(\sum_{\substack{k \in m_1 \\ k \text{ in group } g}} \text{Area}(\text{Int}(j,k)) \right) \quad \text{for all polygons } j,k \text{ and groups } g$$

  where Int is a function that takes two polygons and returns the resulting polygon from their intersection (could be empty).

- The increase in area across a group, C_g is calculated simply as

$$C_g = A_{\text{closing},g} - B_g$$

  and similarly the decrease D_g

$$D_g = B_g - A_{\text{opening},g}$$

- Then the net change across a group E_g is

$$E_g = A_{\text{closing},g} - A_{\text{opening},g} \qquad \text{or} \qquad E_g = C_g + D_g$$

See below an extent account corresponding to the above map.

	231	311	511	512	Total
opening	11.50	108.67	5.44	2.60	128.21
increase	1.12	12.77	0.00	1.27	15.16
decrease	-6.60	-1.28	-5.44	-1.85	-15.16
net change	-5.47	11.49	-5.44	-0.58	0.00
closing	6.03	120.16	0.00	2.02	128.21

We can also represent the increase, decrease and net change as a percent of the opening area.

	231	311	511	512	Total
1	0.10	0.12	0.00	0.49	0.12
2	-0.57	-0.01	-1.00	-0.71	-0.12
3	-0.48	0.11	-1.00	-0.22	0.00

By a similar method to making the extent account above, we can construct an Ecosystem Type Change Matrix to see which codes were added to or taken away from other codes between opening and closing. The diagonal entries are the areas of the unchanged portions. The off-diagonals are the amounts changed from the type in the row to the type in the column.

For example, the Opening extent for (habitat) code *231* was 11.50 Ha. With 6.59 Ha lost to code 311 and 1.12 Ha gained from code 311 also, the net decrease of 5.47 Ha results in a closing extent of 6.03 Ha.

	231	311	511	512	openings
231	4.90	6.59	0.00	0.01	11.50
311	1.12	107.39	0.00	0.16	108.67
511	0.00	4.34	0.00	1.10	5.44
512	0.00	1.85	0.00	0.75	2.60
closings	6.03	120.16	0.00	2.02	128.21

R Code

The maps were generated using [1, 2, 3, 4] and then saved using [5, 6, 7]

The code for this is in the preliminary stages of a shiny app for users with map data of a site over 2 time points to explore the changes across some grouping variable [8].

Makefile

To help with running the R code repeatedly to generate the plots and tables seen in this document, I made use of a Makefile, based on Rob J Hyndman's Blog on the subject[9].

References

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[2] Edzer Pebesma. "Simple Features for R: Standardized Support for Spatial Vector Data". In: *The R Journal* 10.1 (2018), pp. 439–446. DOI: `10.32614/RJ-2018-009`. URL: `https://doi.org/10.32614/RJ-2018-009`.

[3] Edzer Pebesma and Roger Bivand. *Spatial Data Science: With applications in R*. Chapman and Hall/CRC, 2023, p. 352. URL: `https://r-spatial.org/book/`.

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[6] Tim Appelhans et al. *mapview: Interactive Viewing of Spatial Data in R*. R package version 2.11.0.9006. 2023. URL: `https://github.com/r-spatial/mapview`.

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[8] Winston Chang et al. *shiny: Web Application Framework for R*. R package version 1.7.4. 2022. URL: `https://CRAN.R-project.org/package=shiny`.

[9] Rob J Hyndman. *Makefiles for R/LaTeX projects*. Accessed on 21/05/2023. 2012. URL: `https://robjhyndman.com/hyndsight/makefiles/`.